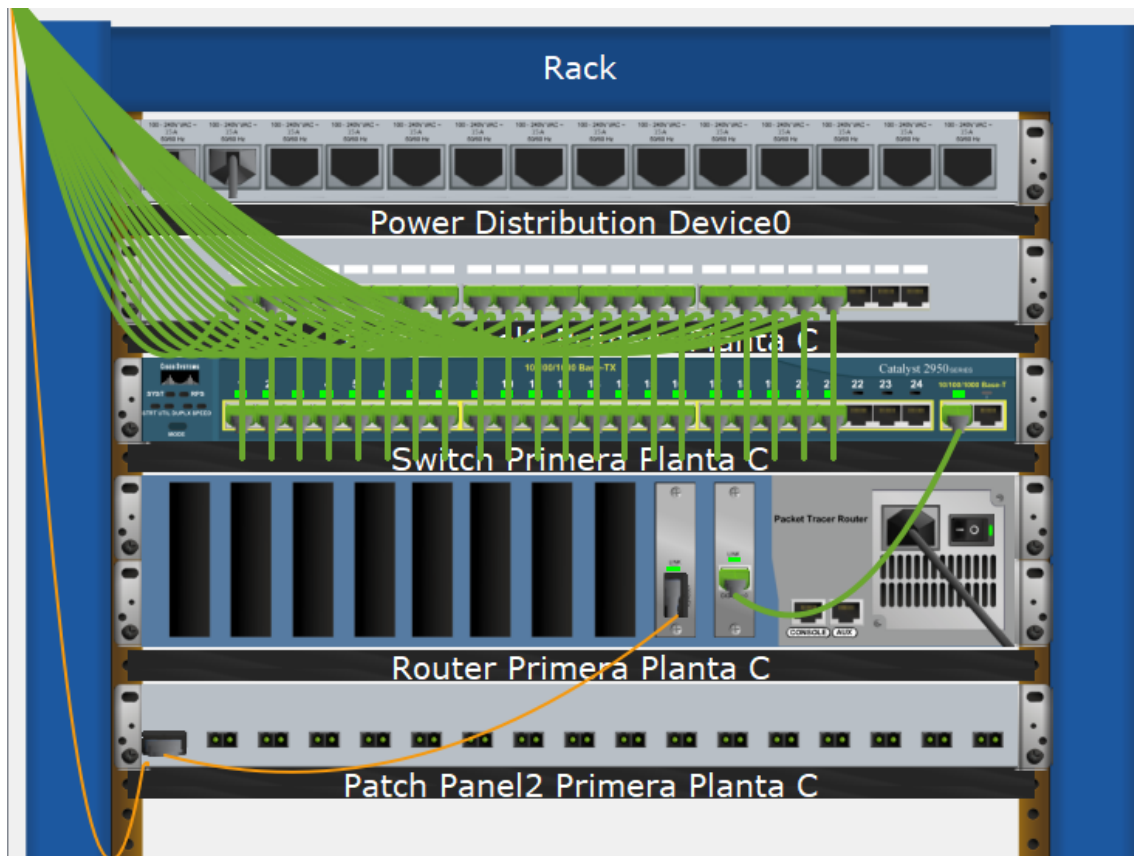
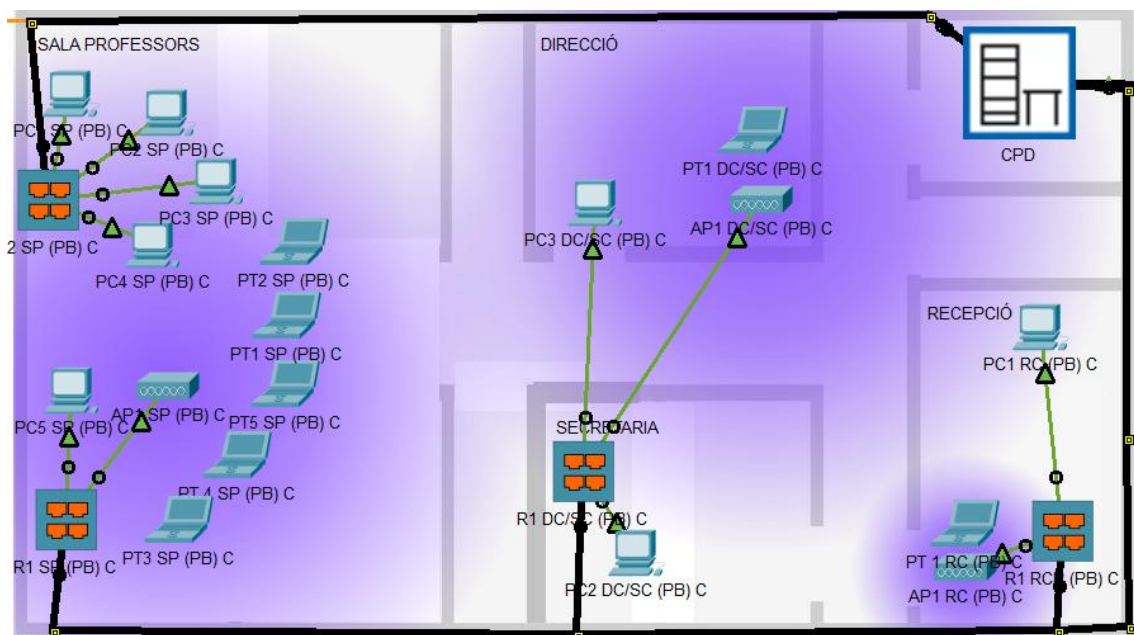
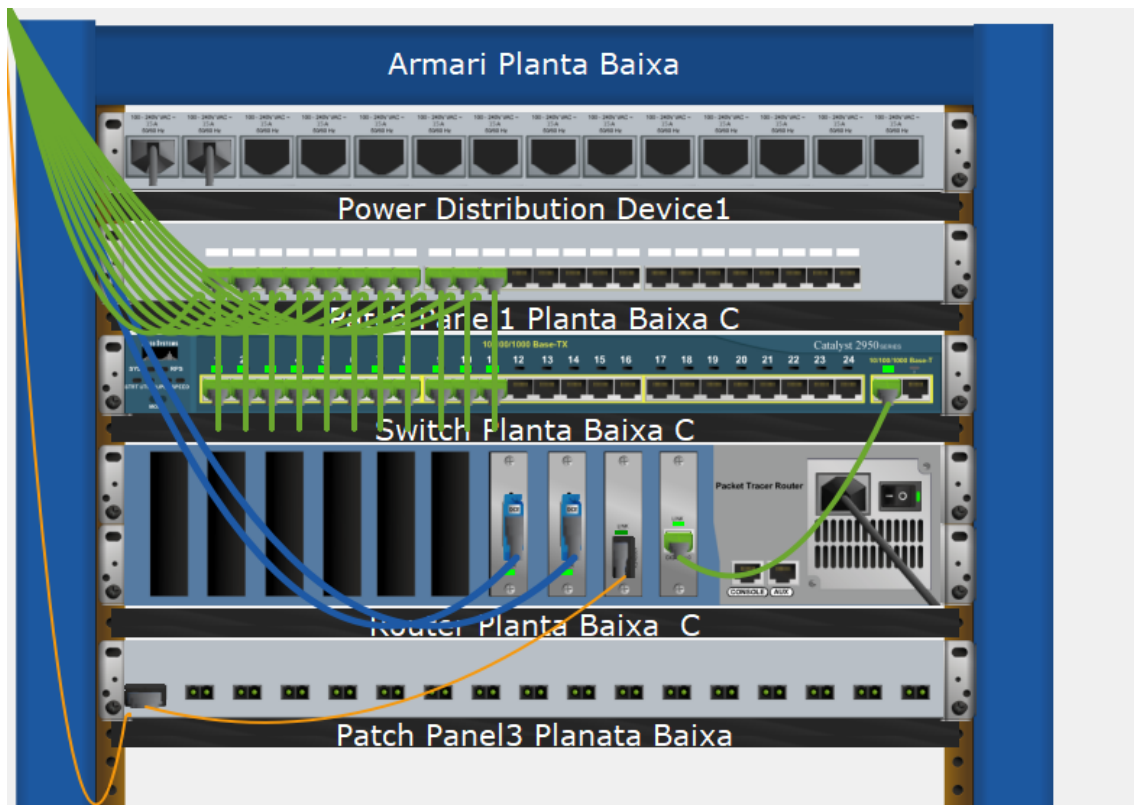




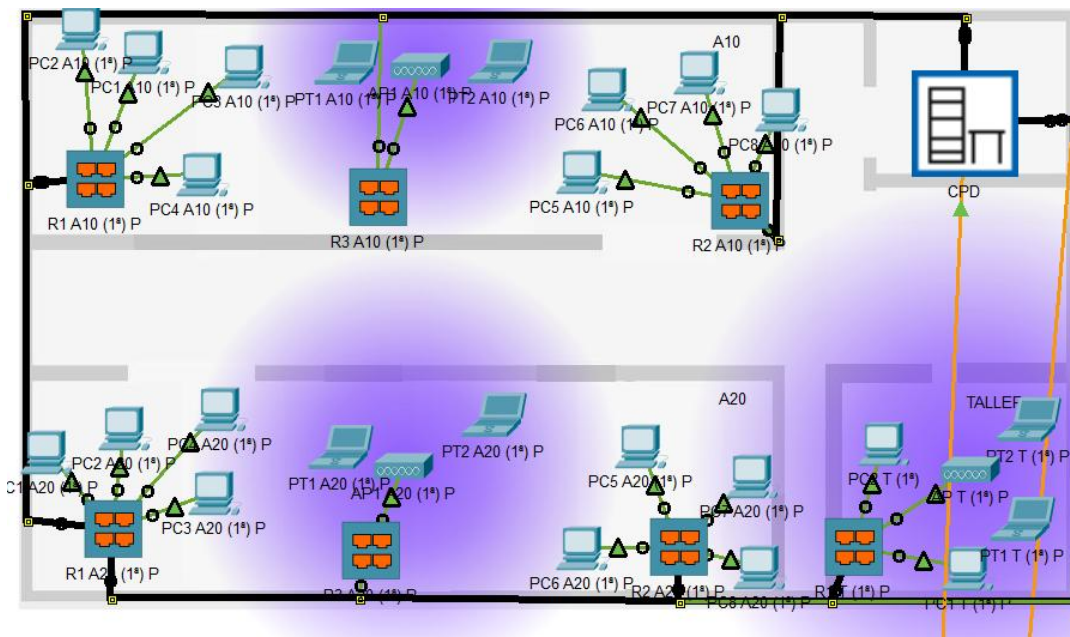
Planta baja córrega



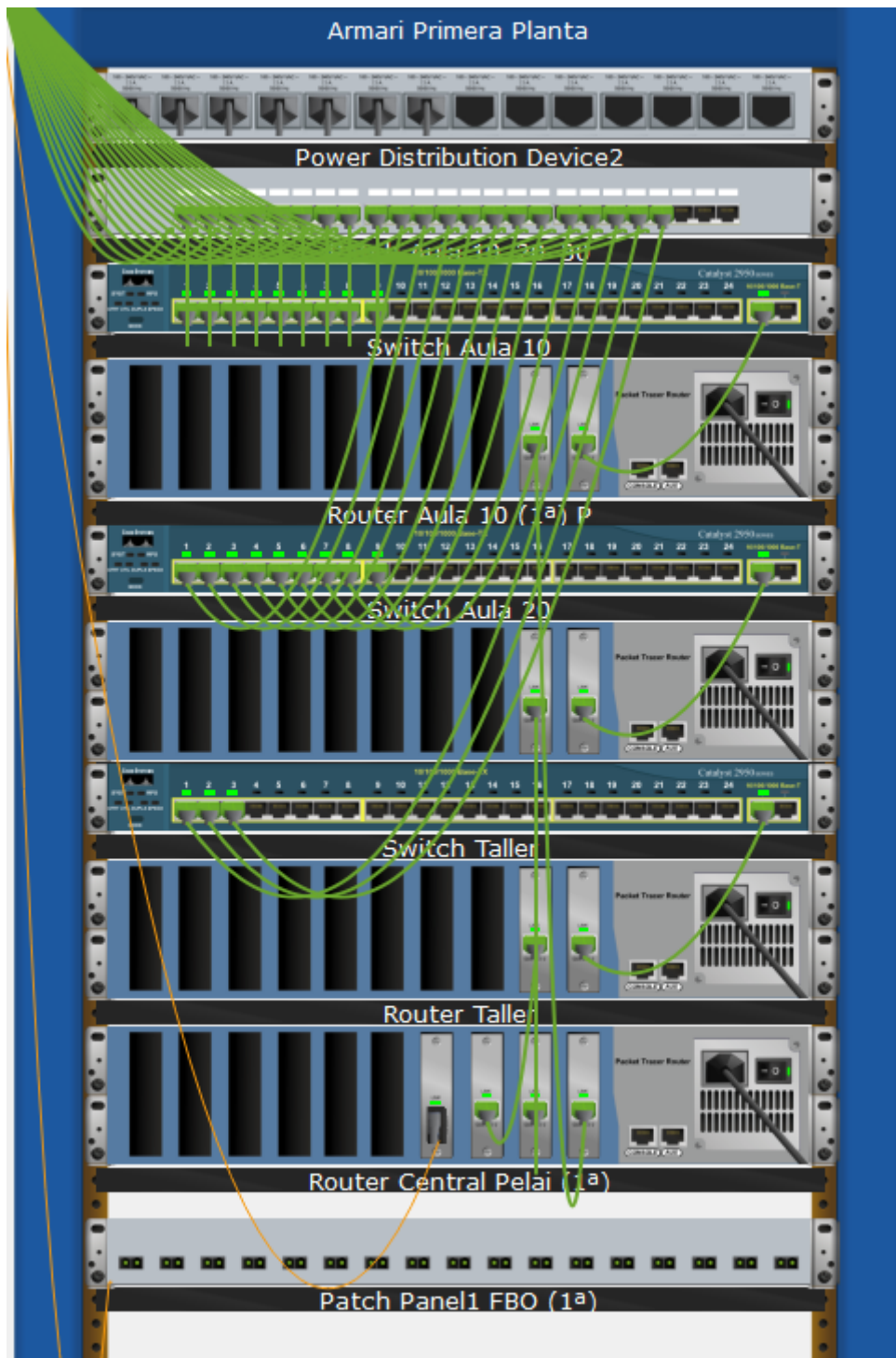
Armario planta baja córrega:



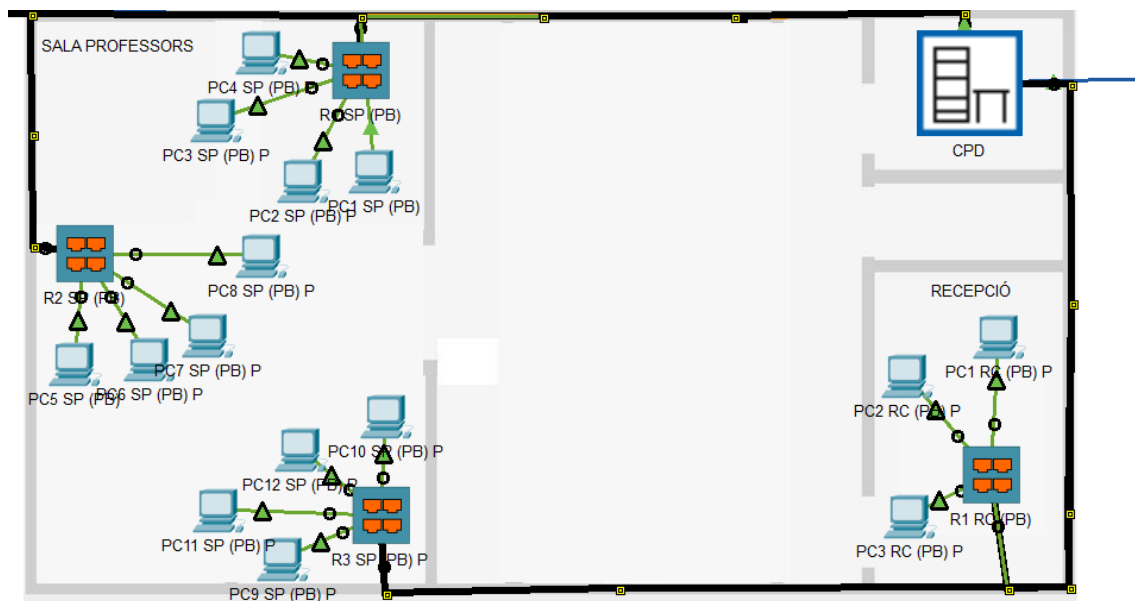
Primera planta pelayo:



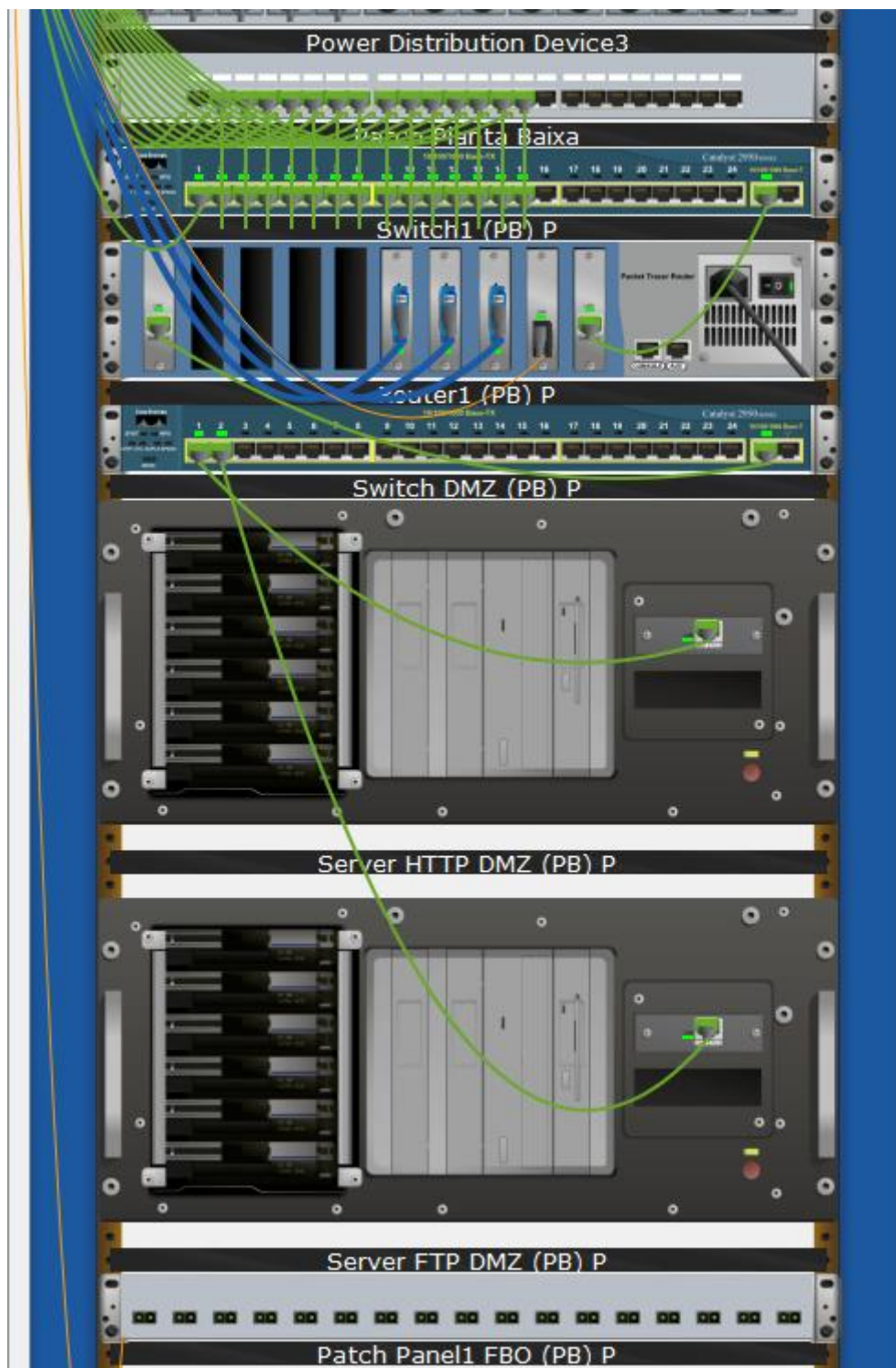
Armario primera planta pelayo:



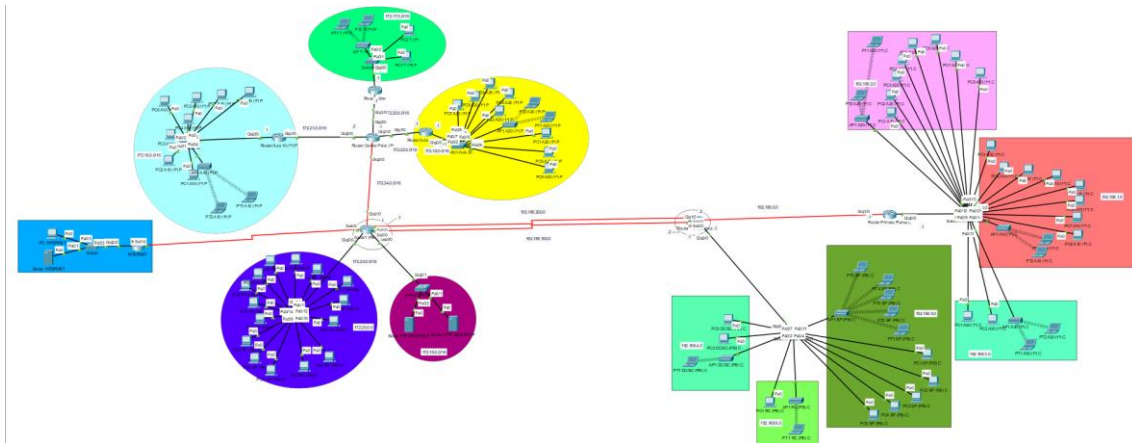
Planta baja pelayo:



Armario planta baja pelayo:



Plano lógico de la red de la escuela.



Documentacion sobre la configuración física de un switch y un wap.

Configuración switch planta baja pelayo:

```
interface GigabitEthernet0/1
  switchport trunk allowed vlan 2-1001
  switchport mode trunk
```

```

interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/10
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/11
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 20
 switchport mode access

```

Configuración acces point primera planta pelayo:

Port 1

Port Status

☒ On

SSID

Default

2.4 GHz Channel

1

Coverage Range (meters)

5,00

Authentication

☐ Disabled

☐ WEP

☒ WPA2-PSK

WEP Key

PSK Pass Phrase

hola1234

User ID

Password

Encryption Type

AES

AP T (1ª) P

Physical

Config

Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status

☒ On

SSID

Default

2.4 GHz Channel

6

Coverage Range (meters)

8,00

Authentication

☐ Disabled

☐ WEP

☒ WPA2-PSK

WEP Key

PSK Pass Phrase

hola1234

User ID

Password

Encryption Type

AES

Port 1

Port Status

☒ On

SSID

Default

2.4 GHz Channel

11

Coverage Range (meters)

6,10

Authentication

☐ Disabled

☐ WEP

☒ WPA2-PSK

WEP Key

PSK Pass Phrase

hola1234

User ID

Password

Encryption Type

AES

Configuración switch primera planta Córcega

```
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/10
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/11
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/12
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/13
 switchport access vlan 20
 switchport mode access
```

```
interface FastEthernet0/14
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/15
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/16
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/17
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/18
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/19
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/20
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/21
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
 switchport trunk allowed vlan 2-1001
 switchport mode trunk
!
interface GigabitEthernet0/2
 switchport trunk allowed vlan 2-1001
 switchport mode trunk
!
```

Configuración switch planta baja:

```

interface FastEthernet0/1
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/2
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/4
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/5
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/6
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/7
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/8
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/9
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/10
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/11
  switchport access vlan 20
  switchport mode access

interface GigabitEthernet0/1
  switchport access vlan 20
  switchport trunk allowed vlan 2-1001
  switchport mode trunk

```

Access Point primera planta corcega:

AP1 A10 (1ª) C

Physical **Config** Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status ☒ On

SSID Default

2.4 GHz Channel 1

Coverage Range (meters) 4,00

Authentication

☐ Disabled ☐ WEP ☒ WPA2-PSK

WEP Key

PSK Pass Phrase hola1234

User ID

Password

Encryption Type AES

AP1 A30 (1ª) C

Physical **Config** Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status ☒ On

SSID Default

2.4 GHz Channel 11

Coverage Range (meters) 4,00

Authentication

☐ Disabled ☐ WEP ☒ WPA2-PSK

WEP Key

PSK Pass Phrase hola1234

User ID

Password

Encryption Type AES

AP1 A20 (1ª) C

Physical **Config** Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status ☒ On

SSID Default

2.4 GHz Channel 6

Coverage Range (meters) 4,00

Authentication

☐ Disabled ☐ WEP ☒ WPA2-PSK

WEP Key

PSK Pass Phrase hola1234

User ID

Password

Encryption Type AES

Configuración Access Point planta baja Corcega:

AP1 SP (PB) C

Physical

Config

Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status

On

SSID

Default

5 GHz Channel

36

Coverage Range (meters)

10,00

Authentication

Disabled

WPA-PSK

WEP

WPA2-PSK

WEP Key

PSK Pass Phrase

hola1234

User ID

Password

Encryption Type

AES

Top

AP1 RC (PB) C

Physical

Config

Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status

On

SSID

Default

5 GHz Channel

64

Coverage Range (meters)

3,05

Authentication

Disabled

WPA-PSK

WEP

WPA2-PSK

WEP Key

PSK Pass Phrase

hola1234

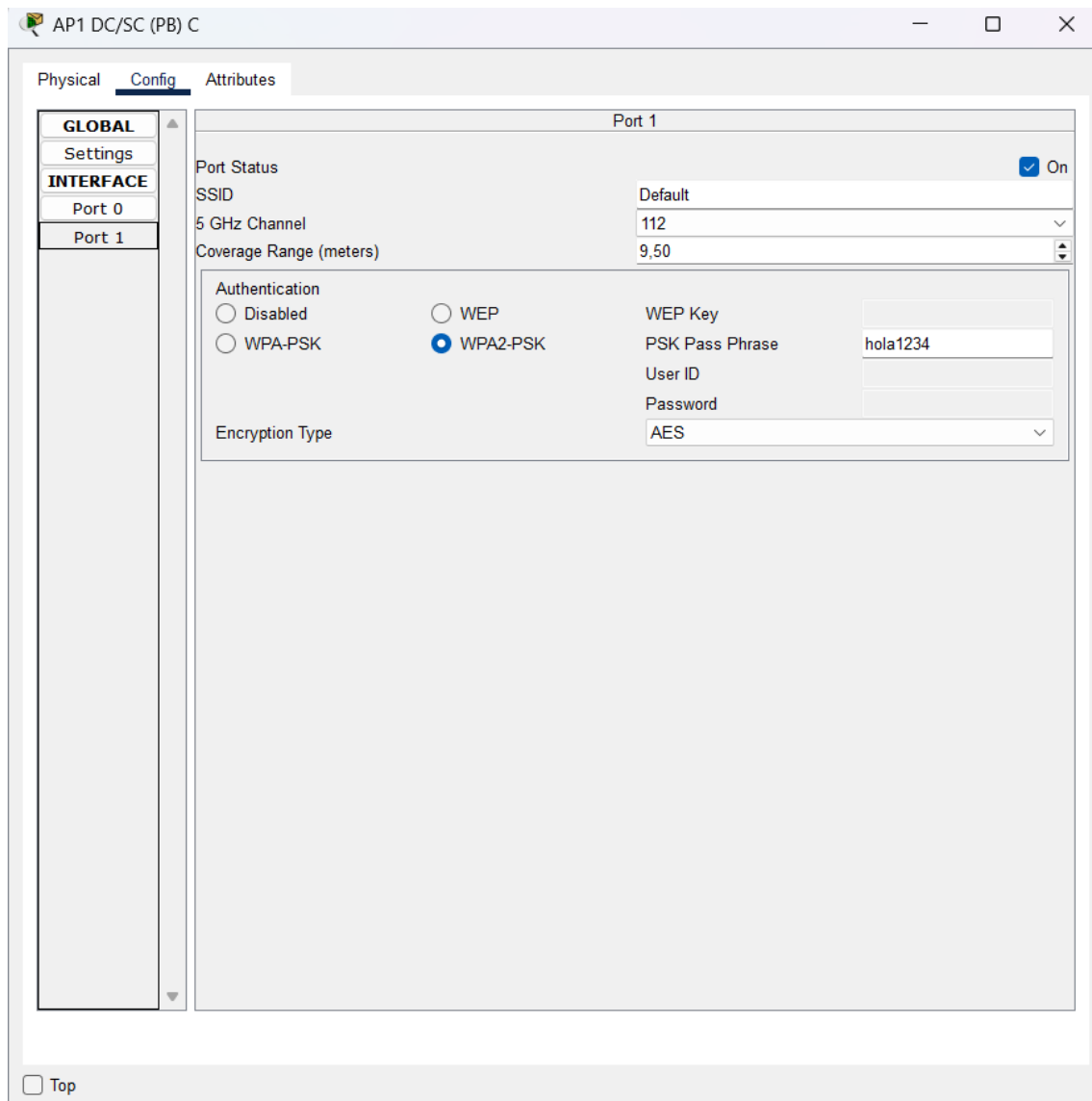
User ID

Password

Encryption Type

AES

Top



Pruebas de connexion:

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time=3ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Prueba conexion corcega primera planta

```
C:\>ping 192.168.6.2

Pinging 192.168.6.2 with 32 bytes of data:

Reply from 192.168.6.2: bytes=32 time<1ms TTL=127
Reply from 192.168.6.2: bytes=32 time=11ms TTL=127
Reply from 192.168.6.2: bytes=32 time<1ms TTL=127
Reply from 192.168.6.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.6.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>ping 192.168.5.2

Pinging 192.168.5.2 with 32 bytes of data:

Reply from 192.168.5.2: bytes=32 time=19ms TTL=127
Reply from 192.168.5.2: bytes=32 time=18ms TTL=127
Reply from 192.168.5.2: bytes=32 time=3ms TTL=127
Reply from 192.168.5.2: bytes=32 time=11ms TTL=127

Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 19ms, Average = 12ms
```

prueba conexion corcega planta baja

```
C:\>ping 172.17.0.2

Pinging 172.17.0.2 with 32 bytes of data:

Reply from 172.17.0.2: bytes=32 time<1ms TTL=125
Reply from 172.17.0.2: bytes=32 time<1ms TTL=125
Reply from 172.17.0.2: bytes=32 time<1ms TTL=125
Reply from 172.17.0.2: bytes=32 time=12ms TTL=125

Ping statistics for 172.17.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 172.18.0.2

Pinging 172.18.0.2 with 32 bytes of data:

Reply from 172.18.0.2: bytes=32 time<1ms TTL=125
Reply from 172.18.0.2: bytes=32 time<1ms TTL=125
Reply from 172.18.0.2: bytes=32 time<1ms TTL=125
Reply from 172.18.0.2: bytes=32 time<1ms TTL=125

Ping statistics for 172.18.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.19.0.2

Pinging 172.19.0.2 with 32 bytes of data:

Reply from 172.19.0.2: bytes=32 time=10ms TTL=125
Reply from 172.19.0.2: bytes=32 time<1ms TTL=125
Reply from 172.19.0.2: bytes=32 time<1ms TTL=125
Reply from 172.19.0.2: bytes=32 time<1ms TTL=125

Ping statistics for 172.19.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Prueba conexion Pelayo

```
C:\>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:

Reply from 192.168.100.2: bytes=32 time=1ms TTL=252
Reply from 192.168.100.2: bytes=32 time=2ms TTL=252
Reply from 192.168.100.2: bytes=32 time=2ms TTL=252
Reply from 192.168.100.2: bytes=32 time=1ms TTL=252

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>ping 80.0.0.3

Pinging 80.0.0.3 with 32 bytes of data:

Reply from 80.0.0.3: bytes=32 time=1ms TTL=124
Reply from 80.0.0.3: bytes=32 time=1ms TTL=124
Reply from 80.0.0.3: bytes=32 time=1ms TTL=124
Reply from 80.0.0.3: bytes=32 time=1ms TTL=124

Ping statistics for 80.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

Prueba conexión pelayo hacia internet y hacia router Córcega

Configuración routers:

Routers primera planta pelayo:

```
interface GigabitEthernet0/0
ip address 172.17.0.1 255.255.0.0
duplex auto
speed auto
!
interface GigabitEthernet1/0
ip address 172.20.0.1 255.255.0.0
duplex auto
speed auto
!
ip classless
ip route 172.16.0.0 255.255.0.0 172.20.0.2
ip route 172.18.0.0 255.255.0.0 172.20.0.2
ip route 172.19.0.0 255.255.0.0 172.20.0.2
ip route 172.24.0.0 255.255.0.0 172.20.0.2
ip route 172.22.0.0 255.255.0.0 172.20.0.2
ip route 172.23.0.0 255.255.0.0 172.20.0.2
ip route 172.25.0.0 255.255.0.0 172.20.0.2
!
```

Esta es la configuración del router del taller de pelayo

```
interface GigabitEthernet0/0
ip address 172.16.0.1 255.255.0.0
duplex auto
speed auto
!
interface GigabitEthernet1/0
ip address 172.21.0.1 255.255.0.0
duplex auto
speed auto
!
ip classless
ip route 172.17.0.0 255.255.0.0 172.21.0.2
ip route 172.18.0.0 255.255.0.0 172.21.0.2
ip route 172.24.0.0 255.255.0.0 172.21.0.2
ip route 172.19.0.0 255.255.0.0 172.21.0.2
ip route 172.23.0.0 255.255.0.0 172.21.0.2
ip route 172.25.0.0 255.255.0.0 172.21.0.2
ip route 172.22.0.0 255.255.0.0 172.21.0.2
ip route 172.20.0.0 255.255.0.0 172.21.0.2
ip route 0.0.0.0 0.0.0.0 172.21.0.2
!
```

Esta es la configuración del router del aula 10 de pelayo

```
interface GigabitEthernet0/0
ip address 172.18.0.1 255.255.0.0
duplex auto
speed auto
!
interface GigabitEthernet1/0
ip address 172.22.0.1 255.255.0.0
duplex auto
speed auto
!
ip classless
ip route 172.16.0.0 255.255.0.0 172.22.0.2
ip route 172.17.0.0 255.255.0.0 172.22.0.2
ip route 172.21.0.0 255.255.0.0 172.22.0.2
ip route 172.24.0.0 255.255.0.0 172.22.0.2
ip route 172.19.0.0 255.255.0.0 172.22.0.2
ip route 172.23.0.0 255.255.0.0 172.22.0.2
ip route 172.25.0.0 255.255.0.0 172.22.0.2
!
```

Esta es la configuración del router del aula 20 de pelayo

```
interface GigabitEthernet0/0
ip address 172.21.0.2 255.255.0.0
duplex auto
speed auto
!
interface GigabitEthernet1/0
ip address 172.22.0.2 255.255.0.0
duplex auto
speed auto
!
interface GigabitEthernet2/0
ip address 172.20.0.2 255.255.0.0
duplex auto
speed auto
!
interface GigabitEthernet3/0
ip address 172.24.0.2 255.255.0.0
!
ip classless
ip route 172.16.0.0 255.255.0.0 172.21.0.1
ip route 172.18.0.0 255.255.0.0 172.22.0.1
ip route 172.19.0.0 255.255.0.0 172.24.0.1
ip route 172.23.0.0 255.255.0.0 172.24.0.1
ip route 172.25.0.0 255.255.0.0 172.24.0.1
ip route 172.17.0.0 255.255.0.0 172.20.0.1
ip route 0.0.0.0 0.0.0.0 172.24.0.1
```

Esta es la configuración del router central de pelayo

Configuracion resto de routers de pelayo:

```
interface GigabitEthernet0/0.1
 encapsulation dot1q 10
 ip address 172.23.0.1 255.255.0.0
!
interface GigabitEthernet0/0.2
 encapsulation dot1q 20
 ip address 172.23.0.2 255.255.0.0
!
interface GigabitEthernet0/0.3
 encapsulation dot1q 30
 no ip address
!
interface Serial0/0/0
 ip address 172.24.0.1 255.255.0.0
!
interface Serial0/0/1
 ip address 192.168.200.1 255.255.255.0
 encapsulation ppp
!
interface Serial0/0/2
 ip address 10.1.1.100.1 255.255.255.0
 encapsulation ppp
!
interface Serial0/0/3
 ip address 212.145.151.2 255.255.255.0
!
interface GigabitEthernet0/0
 ip address 172.19.0.1 255.255.0.0
 duplex auto
 speed auto
!
ip classless
ip route 172.16.0.0 255.255.0.0 0.0.0.0 172.24.0.2
ip route 172.17.0.0 255.255.0.0 0.0.0.0 172.24.0.2
ip route 172.18.0.0 255.255.0.0 0.0.0.0 172.24.0.2
ip route 172.19.0.0 255.255.0.0 0.0.0.0 172.24.0.2
ip route 172.20.0.0 255.255.0.0 0.0.0.0 172.24.0.2
ip route 172.21.0.0 255.255.0.0 0.0.0.0 172.24.0.2
ip route 172.22.0.0 255.255.0.0 0.0.0.0 172.24.0.2
ip route 172.23.0.0 255.255.0.0 0.0.0.0 172.24.0.2
ip route 0.0.0.0 0.0.0.0 0.0.0.0 212.145.151.1
ip route 192.168.0.0 255.255.255.0 192.168.100.2
ip route 192.168.10.0 255.255.255.0 192.168.200.2
ip route 192.168.11.0 255.255.255.0 192.168.200.2
ip route 192.168.12.0 255.255.255.0 192.168.200.2
ip route 192.168.13.0 255.255.255.0 192.168.200.2
ip route 192.168.14.0 255.255.255.0 192.168.200.2
ip route 192.168.15.0 255.255.255.0 192.168.200.2
ip route 192.168.16.0 255.255.255.0 192.168.200.2
ip route 192.168.17.0 255.255.255.0 192.168.200.2
ip route 192.168.18.0 255.255.255.0 192.168.200.2
ip route 192.168.19.0 255.255.255.0 192.168.200.2
ip route 192.168.20.0 255.255.255.0 192.168.200.2
```

Esta es la configuración del router que conecta las dos plantas de pelayo y conecta el edificio de Córcega e internet

Configuracion router internet

```
interface GigabitEthernet0/0
 ip address 80.0.0.1 255.0.0.0
 duplex auto
 speed auto
!
interface Serial1/0
 ip address 212.145.151.1 255.255.255.0
 clock rate 4000000
!
ip classless
ip route 0.0.0.0 0.0.0.0 212.145.151.2
```

Esta es la configuración del router de internet

Configuracion routers Corcega

```

interface GigabitEthernet0/0
  no ip address
  duplex auto
  speed auto
!
interface GigabitEthernet0/0.1
  encapsulation dot1Q 10
  ip address 192.168.1.1 255.255.255.0
!
interface GigabitEthernet0/0.2
  encapsulation dot1Q 20
  ip address 192.168.2.1 255.255.255.0
!
interface GigabitEthernet0/0.3
  encapsulation dot1Q 30
  ip address 192.168.3.1 255.255.255.0
!
interface GigabitEthernet1/0
  ip address 192.168.0.1 255.255.255.0
!
ip classless
ip route 0.0.0.0 0.0.0.0 192.168.0.2

```

Esta es la configuración del router de la primera planta de corcega

```

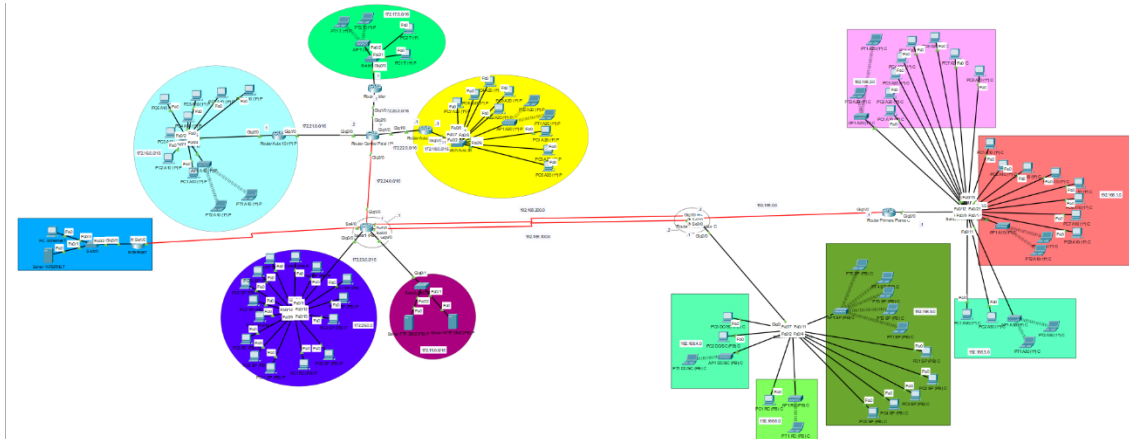
interface GigabitEthernet0/0
  no ip address
  duplex auto
  speed auto
!
interface GigabitEthernet0/0.1
  encapsulation dot1Q 10
  ip address 192.168.1.1 255.255.255.0
!
interface GigabitEthernet0/0.2
  encapsulation dot1Q 20
  ip address 192.168.2.1 255.255.255.0
!
interface GigabitEthernet0/0.3
  encapsulation dot1Q 30
  ip address 192.168.3.1 255.255.255.0
!
interface GigabitEthernet1/0
  ip address 192.168.0.2 255.255.255.0
!
interface Serial2/0
  ip address 192.168.200.2 255.255.255.0
  encapsulation ppp
  clock rate 2000000
!
interface Serial3/0
  ip address 192.168.200.2 255.255.255.0
  encapsulation ppp
  clock rate 2000000
!
router rip
!
ip classless
ip route 192.168.1.0 255.255.255.0 192.168.0.1
ip route 192.168.2.0 255.255.255.0 192.168.0.1
ip route 192.168.3.0 255.255.255.0 192.168.0.1
ip route 0.0.0.0 0.0.0.0 192.168.200.1
ip route 0.0.0.0 0.0.0.0 192.168.100.1

```

Esta es la configuración del router que conecta una planta de corcega con otra y también conecta con el edificio de paylo

Diagrama de red:

Cada Aula tiene su propia red en el edificio de paylo tenemos las redes de 172.16.0.0 hasta 172.25.0.0, en el



edificio de corcega tenemos las redes 192.168.0.0 hasta 192.168.6.0 para conectar ambos edificios usamos dos redes distintas que son 192.168.100.0 y 192.168.200.0 y así haya conexión entre las dos plantas